An 18-Vertex Counterexample to WOWII Conjecture 59

A reproducible counterexample note

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Abstract

We give an explicit connected simple graph G on 18 vertices for which

$$\mathrm{residue}(G) = 10, \qquad b(G) = 17, \qquad f(G) = 13,$$

where $b(G)$ is the maximum order of an induced bipartite subgraph and $f(G)$ is the maximum order of an induced forest. Consequently,

$$\left\lceil \sqrt{\mathrm{residue}(G) b(G)} \right\rceil = \left\lceil \sqrt{170} \right\rceil = 14 > 13 = f(G),$$

disproving the stated form of Written on the Wall II, Conjecture 59. The argument below proves all three invariant values exactly. An accompanying standard-library Python program independently enumerates every one of the 2^{18} induced vertex subsets.

1 The conjectured inequality

The form of WOWII Conjecture 59 considered here is

$$f(G) \;\geq\; \left\lceil \sqrt{\mathrm{residue}(G) b(G)} \right\rceil \tag{1}$$

for every finite simple connected graph G. Here:

- $f(G)$ is the maximum number of vertices in an induced forest of G;
- $b(G)$ is the maximum number of vertices in an induced bipartite subgraph of G; and
- $\mathrm{residue}(G)$ is the Havel–Hakimi residue, namely the number of zeros left when the degree sequence is reduced until all remaining entries are zero.

Theorem 1. *There exists a connected graph G on 18 vertices with*

$$\mathrm{residue}(G) = 10, \qquad b(G) = 17, \qquad f(G) = 13.$$

In particular, G violates (1).

2 Construction of the graph

Let

$$A = \{a_0, a_1, a_2, a_3, a_4\}, \qquad B = \{b_0, b_1, b_2, b_3, b_4\},$$

let h be one additional vertex, and let

$$L = \{\ell_1, \ell_2, \dots, \ell_7\}$$

be seven further vertices. Thus

$$V(G) = A \cup B \cup \{h\} \cup L.$$

The graph induced by $A \cup B$ is bipartite. Its neighborhoods on the B side are given in Table 1.

Table 1: Neighborhoods in the ten-vertex bipartite core.

Vertex	Neighborhood in A	Degree in the core
b_0	$\{a_0, a_1\}$	2
b_1	$\{a_0, a_1, a_2\}$	3
b_2	$\{a_0, a_1, a_2, a_3\}$	4
b_3	A	5
b_4	A	5

Finally, join h to every other vertex and add no further edges. In particular, each ℓ_i is a pendant vertex adjacent only to h. Since h is universal, G is connected. The core has 19 edges and the universal hub adds 17, so $|E(G)| = 36$.

For machine verification we use the numerical labeling

$$a_i = i, \qquad b_i = 5 + i, \qquad h = 10, \qquad L = \{11, 12, 13, 14, 15, 16, 17\}.$$

3 The induced bipartite number

Proposition 1. *The graph G satisfies $b(G) = 17$.*

Proof. Deleting h leaves the bipartite core on $A \cup B$ together with seven isolated vertices. Hence $G - h$ is an induced bipartite subgraph on 17 vertices, and therefore $b(G) \geq 17$.

On the other hand, a_0b_0, b_0h, and ha_0 are all edges. Thus $\{h, a_0, b_0\}$ spans a triangle, so the full 18-vertex graph is not bipartite. No induced bipartite subgraph can therefore have order 18. Hence $b(G) = 17$. □

4 The induced forest number

Write $H = G[A \cup B]$ for the ten-vertex bipartite core.

Lemma 1. *The independence number of H is* 5.

Proof. The part A is independent, so $\alpha(H) \geq 5$. The five edges

$$a_0b_0, \quad a_1b_1, \quad a_2b_2, \quad a_3b_3, \quad a_4b_4$$

form a perfect matching of H. An independent set contains at most one endpoint from each matching edge, so it has size at most 5. Therefore $\alpha(H) = 5$. □

Lemma 2. *Every set of seven vertices of H contains a 4-cycle.*

Proof. Let $S \subseteq V(H)$ have $|S| = 7$, and put

$$a = |S \cap A|, \qquad b = |S \cap B|.$$

Since $a + b = 7$ and $a, b \leq 5$, the possibilities are

$$(a, b) \in \{(5, 2), (4, 3), (3, 4), (2, 5)\}\,.$$

The neighborhoods in Table 1 are nested:

$$N(b_0) \subset N(b_1) \subset N(b_2) \subset N(b_3) = N(b_4) = A,$$

with sizes $2, 3, 4, 5, 5$.

(a) If $(a, b) = (5, 2)$, the two selected vertices of B have at least two common neighbors in A.

(b) If $(a, b) = (4, 3)$, choose the two selected B-vertices with largest neighborhoods. Their common neighborhood has size at least 3. Since only one vertex of A is omitted, at least two of those common neighbors lie in S.

(c) If $(a, b) = (3, 4)$, the two selected B-vertices with largest neighborhoods have at least four common neighbors in A. The complement of that common neighborhood inside A has size at most one, so any selected three vertices of A include at least two common neighbors.

(d) If $(a, b) = (2, 5)$, both b_3 and b_4 are selected and are adjacent to both selected vertices of A.

In every case, two selected vertices of B have two common selected neighbors in A. These four vertices span a $K_{2,2}$ and hence a 4-cycle. □

Proposition 2. *The graph G satisfies $f(G) = 13$.*

Proof. The set

$$B \cup \{h\} \cup L$$

has $5 + 1 + 7 = 13$ vertices and induces a star centered at h. Therefore $f(G) \geq 13$.

For the reverse inequality, let $S \subseteq V(G)$ have $|S| = 14$.

If $h \in S$, then at most seven vertices of $S \setminus \{h\}$ belong to L, so S contains at least six vertices of H. By Lemma 1, those six core vertices contain an edge xy. The universal hub is adjacent to both x and y, so $hxyh$ is a triangle in $G[S]$. Thus $G[S]$ is not a forest.

If $h \notin S$, then at most seven vertices of S belong to L, so S contains at least seven vertices of H. By Lemma 2, those core vertices contain a 4-cycle. Again $G[S]$ is not a forest.

No induced forest has 14 vertices, so $f(G) \leq 13$. Together with the lower bound, this gives $f(G) = 13$. □

5 The Havel–Hakimi residue

The degrees are

- 17 for the hub h;
- $6, 6, 5, 4, 3$ on A;

- $3, 4, 5, 6, 6$ on B; and
- 1 for each of the seven leaves.

Thus the sorted degree sequence is

$$d_0 = [17, 6^4, 5^2, 4^2, 3^2, 1^7],$$

where x^k denotes k copies of x. Successive Havel–Hakimi reductions are

$$\begin{aligned}
d_0 &= [17, 6^4, 5^2, 4^2, 3^2, 1^7], \\
d_1 &= [5^4, 4^2, 3^2, 2^2, 0^7], \\
d_2 &= [4^3, 3^4, 2^2, 0^7], \\
d_3 &= [3^4, 2^4, 0^7], \\
d_4 &= [2^7, 0^7], \\
d_5 &= [2^4, 1^2, 0^7], \\
d_6 &= [2, 1^4, 0^7], \\
d_7 &= [1^2, 0^9], \\
d_8 &= [0^{10}].
\end{aligned}$$

The terminal list has ten zeros, and hence

$$\mathrm{residue}(G) = 10. \tag{2}$$

6 Contradiction to Conjecture 59

Propositions 1 and 2, together with (2), give

$$\mathrm{residue}(G)b(G) = 10 \cdot 17 = 170 = 13^2 + 1.$$

Therefore

$$\left\lceil \sqrt{\mathrm{residue}(G)b(G)} \right\rceil = \left\lceil \sqrt{170} \right\rceil = 14,$$

whereas $f(G) = 13$. The conjectured inequality reads

$$13 \geq 14,$$

which is false. This proves Theorem 1.

Conclusion. The graph above is a connected 18-vertex counterexample to the stated form of WOWII Conjecture 59.

7 Independent exhaustive verification

The repository includes

`verification/verify_counterexample.py`.

The program uses only the Python standard library. It constructs the exact edge set above, checks connectivity, computes the Havel–Hakimi residue, and examines all $2^{18} = 262{,}144$ vertex subsets. Each induced subgraph is tested for bipartiteness by breadth-first two-coloring and for acyclicity by a union–find cycle test. It then asserts the exact values

$$\mathrm{residue}(G) = 10, \qquad b(G) = 17, \qquad f(G) = 13,$$

and the final strict inequality $13 < 14$.

From the repository root, run

```
python3 verification/verify_counterexample.py
```

The expected output is

```
|V| = 18, |E| = 36, connected = True
degree sequence = [17, 6, 6, 6, 6, 5, 5, 4, 4, 3, 3, 1, 1, 1, 1, 1, 1, 1]
Havel--Hakimi residue = 10
b(G) = 17; witness = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17]
f(G) = 13; witness = [0, 1, 2, 3, 4, 5, 11, 12, 13, 14, 15, 16, 17]
ceil(sqrt(residue*b)) = ceil(sqrt(170)) = 14
Conjectured inequality: 13 >= 14  ->  False
```

8 Sources, scope, and attribution

The original WOWII list is available at

http://cms.dt.uh.edu/faculty/delavinae/research/wowII/all.html#conj59.

The corresponding Formal Conjectures transcription is at

https://github.com/google-deepmind/formal-conjectures/blob/main/FormalConjectures/WrittenOnTheWallII/GraphConjecture59.lean.

A related upstream Lean-certification pull request is

https://github.com/google-deepmind/formal-conjectures/pull/4583.

This note concerns the exact inequality and invariant meanings stated in Section 1. It makes no claim of historical or mathematical priority; related upstream work should be consulted when assigning attribution.

A Explicit numeric edge list

Under the labeling

$$A = \{0, 1, 2, 3, 4\}, \quad B = \{5, 6, 7, 8, 9\}, \quad h = 10, \quad L = \{11, 12, 13, 14, 15, 16, 17\},$$

the core edges are

$$\begin{aligned}
&(0, 5), (1, 5),\\
&(0, 6), (1, 6), (2, 6),\\
&(0, 7), (1, 7), (2, 7), (3, 7),\\
&(0, 8), (1, 8), (2, 8), (3, 8), (4, 8),\\
&(0, 9), (1, 9), (2, 9), (3, 9), (4, 9).
\end{aligned}$$

In addition, $(10, v)$ is an edge for every $v \neq 10$. There are no other edges.